Algebraga kirish

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. $x = -2$, $y = 3$ **bo'lsa,** $xy^2 - 2y$ **ning qiymatini toping.**

A) 12 **B)** −24 **C)** 30 **D)** −12

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2. $a = \frac{1}{2}$, $b = -4$ **bo'lsa,** $6a - b^2$ **ning qiymatini toping.**

A) 19 **B)** −5 **C)** 11 **D)** −13

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3. $p = -3$ **bo'lsa,** $-p^2$ **ning qiymatini toping.**

A) −6 **B)** 6 **C)** −9 **D)** 9

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4. $n = -3$ **bo'lsa,** $(2n)^2 - 2n^2$ **ning qiymatini toping.**

A) 54 **B)** −18 **C)** 18 **D)** 0

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5. $y = \frac{n - m^2}{m + n}$ **formulada** $m = -3$, $n = 5$. y **ni toping.**

A) −7 **B)** 7 **C)** 2 **D)** −2

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6. $\frac{1}{x} = \frac{1}{a} - \frac{1}{b}$ **formulada** $a = 3$, $b = 12$. x **ni toping.**

A) $\frac{1}{9}$ **B)** $\frac{1}{4}$ **C)** 4 **D)** 9

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7. $S = \frac{1}{2}n(n+1)$ **— dastlabki** n **ta natural son yig'indisi.** $11 + 12 + \cdots + 30$ **ni toping.**

A) 410 **B)** 399 **C)** 420 **D)** 465

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8. $C = \frac{5}{9}(F - 32)$ **formula Farengeytni Selsiyga o'tkazadi.** $F = -4$ **bo'lsa,** C **ni toping.**

A) −36 **B)** 20 **C)** −4 **D)** −20

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9. **Qisqa yozing:** $4a \cdot 3b \cdot a$

A) $12a^2b$ **B)** $7a^2b$ **C)** $12ab$ **D)** $12a^3b$

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10. **Qisqa yozing:** $18x : 6y \cdot z$

A) $\frac{12xz}{y}$ **B)** $\frac{3x}{yz}$ **C)** $\frac{3xz}{y}$ **D)** $3xyz$

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11. «a **va** b **ning yig'indisi kvadratini** c **ga bo'lish» qanday yoziladi?**

A) $\frac{(a+b)^2}{c}$ **B)** $a + \frac{b^2}{c}$

C) $\frac{a+b}{c^2}$ **D)** $\frac{a^2+b}{c}$

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12. **«y ning qarama-qarshisini x ning kubiga qo'shish» qanday yoziladi?**

A) $x - y^3$ **B)** $x^3 + y$ **C)** $x^3 - y$ **D)** $3x - y$

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13. **Rasmda uchta qopda m tadan, ikkita qutida k tadan kitob bor, yana 4 ta kitob ochiq turibdi. $m = 12$, $k = 9$ bo'lsa, jami nechta kitob bor?**

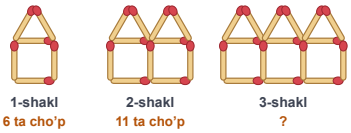


A) 40 **B)** 58

C) 25 **D)** 49

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14. **Rasmdagidek gugurt cho'plaridan qatorasiga uychalar terildi. Jami 76 ta cho'p ishlatildi. Nechta uycha terilgan?**

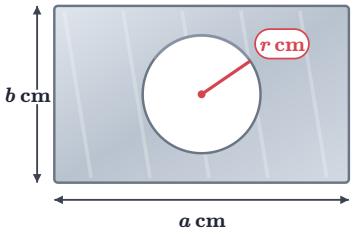


A) 12 **B)** 15

C) 13 **D)** 19

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15. **Rasmdagi metall listdan doira teshik qirqib olindi. $a = 18$, $b = 10$, $r = 3$ bo'lsa, qolgan qismning yuzini toping.**



A) $(180 - 9\pi)$ cm² **B)** $(28 - 9\pi)$ cm²

C) $(180 - 6\pi)$ cm² **D)** 171π cm²

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16. **Avtoturargohda birinchi 2 soat uchun birga 5 000 so'm, undan keyingi har bir soat uchun 3 000 so'm olinadi. Mashina t soat turdi ($t \geq 2$). To'lov qancha?**

A) $5\,000 + 3\,000(t - 2)$ **B)** $8\,000t$

C) $3\,000 + 5\,000(t - 2)$ **D)** $5\,000 + 3\,000t$

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17. **Novvoyxona 4 kun davomida kuniga o'rtacha a ta, keyingi 3 kun kuniga o'rtacha 40 ta tort sotdi. Shu 7 kunda kuniga o'rtacha nechta tort sotilgan?**

A) $\dfrac{4a + 120}{2}$ **B)** $\dfrac{4a + 40}{7}$

C) $\dfrac{4a + 120}{7}$ **D)** $\dfrac{a + 40}{2}$

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18. **To'g'ri to'rtburchak yuzi $S = ab$. Uning eni a 2 marta qisqartirildi, bo'yi b 5 cm uzaytirildi. Yangi yuz qanday yoziladi?**

A) $\dfrac{a}{2}(b + 5)$ **B)** $\dfrac{a}{2}b + 5$

C) $2a(b + 5)$ **D)** $(a - 2)(b + 5)$

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19. **Qaysi vaziyat $50\,000 - 3x$ ifoda bilan yoziladi?**

A) Har biri 50 000 so'mlik 3 ta kitob va x so'mlik daftar narxi

B) x so'mlik kitob 3 marta arzonlashgandagi narx

C) 50 000 so'm bilan har biri x so'mlik 3 ta kitob olingandan keyin qolgan pul

D) 50 000 so'mni 3 kishiga teng bo'lgandagi har birining ulushi

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20. $\frac{n+4}{2}$ **va** $n+2$ **ifodalar teng kuchlimi?**

A) Ha: $n=0$ da ikkalasi ham 2

B) Ha: ikkalasida ham 4 va 2 bor

C) Yo'q: $n=0$ da 2 va 4

D) Yo'q: $n=2$ da 3 va 4

Javoblar

Algebraga kirish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	−24
2	**D**	−13
3	**C**	−9
4	**C**	18
5	**D**	−2
6	**C**	4
7	**A**	410
8	**D**	−20
9	**A**	$12a^2b$
10	**C**	$\dfrac{3xz}{y}$

№	Javob	Qiymati
11	**A**	$\dfrac{(a+b)^2}{c}$
12	**C**	$x^3 - y$
13	**B**	58
14	**B**	15
15	**A**	$(180 - 9\pi)$ cm²
16	**A**	$5\,000 + 3\,000(t-2)$
17	**C**	$\dfrac{4a+120}{7}$
18	**A**	$\dfrac{a}{2}(b+5)$
19	**C**	50 000 so'm bilan har biri x so'mlik 3 ta kitob olingandan keyin qolgan pul
20	**D**	Yo'q: $n = 2$ da 3 va 4

Qisqa yechimlar

1. Darajaga faqat y ko'tariladi: $y^2 = 9$, $xy^2 = (-2)\cdot 9 = -18$.
$2y = 2\cdot 3 = 6$.
$-18 - 6 = -24$. Javob: **−24**. (30 — $(xy)^2 = 36$ deb olinsa.)

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2. $6a = 6\cdot\dfrac{1}{2} = 3$.
$b^2 = (-4)^2 = 16$.
$3 - 16 = -13$. Javob: **−13**. (11 — b^2 o'rniga $2b$ olinsa.)

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3. Avval daraja: $p^2 = (-3)^2 = 9$.
Keyin qarama-qarshisi: -9. Javob: **−9**.

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4. $(2n)^2 = (-6)^2 = 36$.
$2n^2 = 2\cdot 9 = 18$.
$36 - 18 = 18$. Javob: **18**.

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5. Surat: $5 - (-3)^2 = 5 - 9 = -4$.
Maxraj: $-3 + 5 = 2$.
$y = -4 : 2 = -2$. Javob: **−2**. (7 — $-3^2 = -9$ deb olinsa: $(5+9) : 2$.)

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6. $\dfrac{1}{3} - \dfrac{1}{12} = \dfrac{4}{12} - \dfrac{1}{12} = \dfrac{3}{12} = \dfrac{1}{4}$.
$\dfrac{1}{x} = \dfrac{1}{4}$, demak $x = 4$. Javob: **4**. ($\dfrac{1}{4}$ — bu $\dfrac{1}{x}$; 9 — maxrajlar ayirilsa.)

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7. $1 + \cdots + 30 = \dfrac{1}{2}\cdot 30\cdot 31 = 465$.
Ayiriladigani — dastlabki 10 ta son: $\dfrac{1}{2}\cdot 10\cdot 11 = 55$.
$465 - 55 = 410$. Javob: **410**. (399 — 11 ni ham ayirib yuborilsa.)

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8. $F - 32 = -4 - 32 = -36$.
$\frac{5}{9} \cdot (-36) = -20$. Javob: **−20 °C**.

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9. Sonlar: $4 \cdot 3 = 12$.
Harflar: $a \cdot a = a^2$, yana b.
Javob: $12a^2b$.

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10. Chapdan o'ngga: avval $\frac{18x}{6y} = \frac{3x}{y}$.
Keyin z ga ko'paytiramiz: $\frac{3xz}{y}$ (z maxrajga tushmaydi).
Javob: $\frac{3xz}{y}$.

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11. Yig'indi $a + b$ butunligicha kvadratga ko'tariladi: $(a+b)^2$.
Natija c ga bo'linadi. Javob: $\frac{(a+b)^2}{c}$.

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12. x ning kubi — x^3, y ning qarama-qarshisi — $-y$.
$x^3 + (-y) = x^3 - y$. Javob: $x^3 - y$.

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13. Jami: $3m + 2k + 4$.
$3 \cdot 12 + 2 \cdot 9 + 4 = 36 + 18 + 4 = 58$. Javob: **58**.

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14. 1-uycha: 6 ta, 2-uycha: 11 ta — har yangi uychaga 5 ta cho'p qo'shiladi (umumiy devor bir marta): n ta uychaga $5n + 1$ ta cho'p.
$5n + 1 = 76$, $5n = 75$, $n = 15$. Javob: **15**. (12 yoki 13 — har uychani alohida 6 tadan sanash.)

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15. Qolgan yuz: $ab - \pi r^2$.
$18 \cdot 10 - \pi \cdot 3^2 = 180 - 9\pi$. Javob π bilan qoladi: $(180 - 9\pi)$ **cm²** (171π — xato: o'xshash bo'lmagan sonlar birlashtirilmaydi).

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16. Birinchi 2 soat — 5 000 so'm. Qolgan soatlar soni $t - 2$, har biri 3 000 so'm.
Javob: $5\,000 + 3\,000(t - 2)$. ($5\,000 + 3\,000t$ da birinchi 2 soat ikki marta hisoblangan.)

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17. Jami tortlar: $4a + 3 \cdot 40 = 4a + 120$.
O'rtacha = jami : kunlar soni (7).
Javob: $\frac{4a + 120}{7}$. ($\frac{a + 40}{2}$ — kunlar soni har xil ekani hisobga olinmagan.)

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18. Formuladagi a o'rniga $\frac{a}{2}$, b o'rniga $(b + 5)$ qo'yamiz — qavs shart.
Javob: $\frac{a}{2}(b + 5)$.

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19. $3x$ — har biri x so'mlik 3 ta narsa narxi. U 50 000 dan ayirilyapti — qolgan pul.
Javob: **50 000 so'm bilan 3 ta kitob olingandan keyin qolgan pul**.

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20. $n = 0$: $\frac{4}{2} = 2$ va $0 + 2 = 2$ — teng, lekin bitta qiymat yetmaydi.
$n = 2$: $\frac{6}{2} = 3$, $2 + 2 = 4$ — teng emas.
Javob: **yo'q,** $n = 2$ **da 3 va 4**.
